

Agenda Setting through Social Media: The Importance of Incidental News Exposure and Social Filtering in the Digital Era

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Abstract

Conventional models of agenda setting hold that mainstream media influence the public agenda by leading audience attention, and perceived importance, to certain issues. However, increased selectivity and audience fragmentation in today's digital media environment threaten the traditional agenda-setting power of the mass media. An important development to consider in light of this change is the growing use of social media for entertainment and information. This study investigates whether mainstream media can influence the public agenda when channeled through social media. By leveraging an original, longitudinal experiment, I test whether being exposed to political information through Facebook yields an agenda-setting effect by raising participants' perceived importance of certain policy issues. Findings show that participants exposed to political information on Facebook exhibit increased levels of issue salience consistent with the issues shared compared with participants who were not shown political information; these effects are strongest among those with low political interest.

Keywords

agenda setting, social media, Facebook, two-step flow of communication, incidental exposure

During the broadcast era, there were relatively few media outlets. Their programming was able to reach broad audiences, and, therefore, that programming held significant influence over the public agenda. In the present media environment, however, there are far more media sources, allowing for the tailoring of media consumption to suit individual audience members' interests, and thus threatening the long-held ability of the mass media to shape the public agenda (Chaffee and Metzger 2001; McCombs 2005; Prior 2007; Williams and Delli Carpini 2011). This shift from mass broadcasting to large audiences toward niche media reaching more narrowly targeted and attentive audiences is commonly referred to as audience fragmentation and is widely believed to be a source of change in political behavior and public opinion. McCombs (2005, 545) points out that the heterogeneous media available on the Internet, for example, can lead to diverse agendas among the public, a "situation that would spell the demise of agenda setting as we have known it." Subsequently, a public that does not share a common agenda may find it impossible to come together and engage in collective action because its members disagree on what is important to society (Chaffee and Metzger 2001). The fragmentation of audiences, and the

increased ability of those with low levels of political interest to avoid the news agenda altogether (Prior 2007), call into question the continuing ability of the mass media to reach and inform the general public and, therefore, foster consensus on the important issues of the day (Chaffee and Wilson 1977; McCombs and Zhu 1995; Tan and Weaver 2013).

As the audience for mass media contracts, the number of people who report getting their news through social media is growing (Gottfried and Shearer 2016). Social media may engender an agenda-setting effect through the social sharing of political news and, relatedly, through increased incidental exposure to political information among those who might otherwise choose to avoid it. This study asks the following question: can social media convey an agenda-setting effect in an environment marked by abundant media choice and increased individual selectivity? Despite the potential for social media to

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shape the public agenda in the contemporary media environment, significant challenges to the study of these effects have left this question unanswered.

One way in which the mass media agenda is able to reach an audience beyond its direct receivers is through the social filtering of information. This filtering phenomenon was dubbed the “two-step flow of communication” by Katz and Lazarsfeld (1955) who described the process wherein political information is filtered to receivers through the opinion leaders in their personal network. The two-step flow of communication imparts greater personal relevance to political information, which helps to extend the audience for any given media outlet or story and, in doing so, to influence otherwise inattentive citizens. The social sharing of information through social media might help to offset some of the consequences of audience fragmentation, and informational selectivity in particular, and therefore help to distribute the mass media agenda to the broader public.

The social sharing of the media agenda can help information reach a larger audience through incidental exposure. The term “incidental political information” is used to describe exposure to political information that results as a second-order effect of first-order entertainment seeking. However, recent technological developments, such as user-curated content and news feeds, make it easy for receivers to customize their media consumption, resulting in less exposure to incidental political information in the digital media environment. The overall reduction of incidental exposure to political information not only leads to lower levels of political knowledge and participation (Prior 2007), but also poses an additional threat to the traditional agenda-setting power of the mass media in the form of personalized political agendas (Chaffee and Metzger 2001; McCombs 2005).

In this article, I argue that social media can serve an agenda-setting function by providing users with incidental political information as filtered through the process of two-step communication flow within their networks. Furthermore, I expect that this agenda-setting effect will be strongest for politically uninterested individuals as they are known to experience a higher level of uncertainty about political affairs (Delli Carpini and Keeter 1996). Uncertainty, coupled with socially imparted issue relevance through the networked sharing of information, makes this population most susceptible to agenda-setting effects conveyed through social media (Matthes 2006; McCombs 2004; McCombs, Shaw, and Weaver 2014).

Many studies of agenda-setting effects are limited in their causal inference and external validity. The present study seeks to overcome these shortcomings by leveraging an original longitudinal experiment administered organically through the Facebook platform. This study uses a randomized controlled design that compares a treatment

group exposed to political information with a control group who were not, and tests whether the treatment group exhibits agenda-setting effects consistent with the information shared. This design improves external validity by exposing participants to a realistic treatment through their own Facebook News Feed administered over the course of nearly three months. The panel data derived from the longitudinal design not only allow for more precise measurement of individual-level change resulting from the treatment (Bartels 2006), but also account for the influence of moderators, such as political interest, on this effect. Overall, the findings show that agenda setting can occur through social media platforms via incidental exposure to political information and that this effect is strongest among the least politically interested. These findings have important implications for the study of agenda-setting effects and for our understanding of how social media can enable more diversified agendas through incidental exposure to political information.

A Fragmented Audience and Agenda Setting

Audience fragmentation, specifically along lines of political interest, raises concerns about the distribution of political information within society. During the broadcast era, even the most politically averse were exposed to some political information—often inadvertently or through their social networks. Prior (2007) demonstrates that public exposure to political information is lower today than during previous decades not only because people are choosing entertainment over news, but also because people are less likely to be inadvertently exposed to political information due to greater selectivity. For example, on-demand TV viewing allows people to see only what they want and to more easily avoid incidental information such as plugs for the nightly news that air before their desired program. Therefore, as fewer people tune in to watch the news, and fewer still encounter incidental exposure as a result of more efficient technology, the ability of the mass media to reach the masses declines. Inadvertent exposure to political information helps to maintain a moderate level of political knowledge among the politically uninterested, but without it, the divide between the information-rich and -poor may endure and even widen (Norris 2001).

Mass media agenda setting and the public agenda have important consequences for policy making and representational accountability. The policy agenda, or the “list of subjects or problems to which governmental officials . . . are paying some serious attention at any given time” (Kingdon 1984, 3) is significantly influenced by the issues that the public perceives to be important. What the public believes is important is heavily influenced by mass

media, which impart issue salience to the general public by choosing to cover certain stories over others (Iyengar and Kinder 1987; McCombs and Shaw 1972; Price and Tewksbury 1997; Scheufele and Tewksbury 2007). Research also demonstrates that the more attention the mass media pay to an issue, the more the public will use that issue in their evaluation of political representatives (Iyengar and Kinder 1987; Krosnick and Kinder 1990). At a basic level, the mass media cover important issues that a responsible public should be aware of and use to evaluate representatives. As the mass media lose their grasp on the public agenda through increased selectivity and audience fragmentation, concerns continue to be raised about the distribution of political information among society, the issues the public perceives to be important, and the criteria to which the public holds their elected officials accountable.

Studies of the agenda-setting power of the media traditionally focus on political campaigns and network news programming (Funkhouser 1973; Iyengar and Kinder 1987; McCombs and Shaw 1972), institutions that, in the digital era, are rapidly changing and arguably less central to many citizens' lives. The audience for network news has been in steady decline since the 1980s as people are increasingly able to select programming that better suits their interests—political or otherwise (Prior 2007; Stroud 2011). Simultaneously, online platforms for news, entertainment, and social networking are growing in popularity. The digital media environment allows for greater media selectivity; this often yields customized information and personalized news (Tewksbury and Rittenberg 2012). Audience fragmentation is further enabled by algorithms that create “filter bubbles,” which limit information exposure based on past searches and interests (Pariser 2011). As a result, the ability of the news media to penetrate these bubbles and communicate a clear agenda to the public is in question; therefore, agenda-setting effects need to be reconsidered in light of abundant choice and rampant audience fragmentation. To what extent can the media influence the public agenda if the public can tailor their own media and avoid news entirely?

Agenda Setting through Social Media

As described previously, the two-step flow theory identifies network opinion leaders as crucial entities that help to distribute mass media messages to otherwise inattentive audiences, thereby facilitating and extending the media's public agenda-setting function (Katz and Lazarsfeld 1955). Recently, however, Bennett and Manheim (2006) challenged the applicability of this theory in the digital media environment. They suggested that the social sharing of mass media through opinion leaders is less likely today because of declining membership in

the social clubs and organizations where opinion leaders are most likely to be found (see also Putnam 2000). The authors noted that social isolation in combination with media targeting and audience selectivity make it more likely that there will be only a one-step flow of information: from the media producer directly to the consumer. If this claim by Bennett and Manheim is accurate, then the mass media agenda would be circumscribed only to the politically attentive public while the inattentive enjoy other programming they prefer more.

While most studies of incidental exposure to political information examine entertainment media (Holbert and Hansen 2006; Holbert et al. 2003; Holbrook and Hill 2005) and soft news (Baum 2002; Baum and Jamison 2006), social media may also convey similar effects. When surveyed, Facebook users rarely reference learning political information as their primary purpose for using the social media platform (Lin and Lu 2011; Quan-Haase and Young 2010). Yet, a 2016 survey by the Pew Research Center found that 66 percent of Facebook users get news from the site and that 62 percent of these users are exposed to news incidentally while doing other things online (Gottfried and Shearer 2016). The majority of social media users log on to connect with their network, socialize, and share personal information; these users are not exhibiting what would commonly be viewed as “news seeking” or “politically interested” behaviors. However, when users open their Facebook or Twitter feeds, they are often exposed to news-quality information just the same.

What do we know about the news and information to which Facebook users are exposed? In an analysis of stories shared over a six-month period, Bakshy, Messing, and Adamic (2015) showed that 13 percent of the stories shared on Facebook were considered “hard content,” featuring national news, politics, or world affairs, and that 7 percent of users clicked on the hard content in their feed. In principle, this suggests that for every ten posts one scrolls through on News Feed, at least one contains hard-news content. This figure is bound to vary at the individual level based on network and algorithmic considerations, however, on average, it represents a nontrivial amount of news shared through a social networking site. While many would contend that social media are increasingly considered to be a primary source for news, the architecture of the Facebook platform, in particular, remains dedicated to social networking—though this may be changing.

The news stories accessible, and therefore shareable online, are highly comparable with those found in “legacy media” (Maier 2010). Recently, and critical to the study of mass media agenda setting through social media, Trilling, Tolochko, and Burscher (2016) demonstrated that there is considerable focus among the stories shared on Facebook when compared with other

social media sites. Topics that are popular in the mass media receive more shares on Facebook than they do on Twitter, suggesting that “sharing on Facebook centers more around a few dominant issues, whereas on Twitter there is more variation” (Trilling, Tolochko, and Burscher 2016, 16). And while some might expect Facebook to facilitate informational echo chambers as a function of the personal networks users assemble, the information shared on Facebook is actually quite diverse (Bakshy et al. 2012). Consequently, if people are exposed to incidental news through social media, and it appears that they are, then social media may convey political information to users and perhaps, therefore, have an agenda-setting effect that reflects the content of mass media.

The Conditional Nature of Agenda Setting and the Importance of Political Interest

Social media may not only enable incidental exposure to political information, but additionally provide a forum through which politically interested people can easily share information among a perhaps less-interested public. The digital sharing of political information, therefore, is similar to the two-step flow of communication that Katz and Lazarsfeld (1955) described in the broadcast era: mass media produce information, opinion leaders share it by posting links on social media, and less attentive citizens enjoy low-cost information shared by the opinion leaders in their social networks. In this way, social media platforms may help to disseminate the mass media agenda to a broader audience through online opinion leaders and incidental exposure, just as in-person opinion leaders did before the Internet.

The role of opinion leaders in the dissemination of information through social media may facilitate agenda-setting effects in particular. The psychological explanation for the transfer of issue salience from the media to the public agenda is determined by one’s “need for orientation” (Matthes 2006; McCombs 2004; McCombs, Shaw, and Weaver 2014; Weaver 1977), in this case, toward politics in general. Need for orientation is a psychological term that describes an innate need to understand and impose structure on the unfamiliar world around us (Matthes 2006, 424; McCombs 2004, 53). One’s need for orientation is derived from two underlying factors: relevance and uncertainty. Relevance is the feeling that an issue is important to society whereas uncertainty accounts for how familiar a person is with a topic. Those with high uncertainty and high relevance have a stronger need for orientation and demonstrate the strongest agenda-setting effects. Relevance and uncertainty have both been operationalized in various ways over the course of study.

Relevance has been operationalized using measures of political interest in elections, discussion of the campaign, and more pointed questions that assess how related a particular issue is to one’s life (Bulkow, Urban, and Schweiger 2013; McCombs and Weaver 1973). McCombs (2004) notes that the mechanism of peer influence conveys additional relevance and is often a reason given by respondents for naming particular problems as “the most important.” The peer conveyance of relevance is particularly important to understanding agenda setting through social media. When a group or an individual shares a news story through social media, they convey both issue salience and relevance simultaneously; this combination makes it likely that the issues encountered through social media, especially from trusted sources, may convey even more salience than if they were encountered through mass media alone.

Uncertainty makes up the second component of one’s need for orientation and has been operationalized using measures of vote consistency over time, strength of party identification, and voter confidence (McCombs and Weaver 1973; Weaver 1980). Based on this logic, agenda-setting effects should be strong among users who have lower levels of political interest because they demonstrate uncertainty and less-stable attitudes and opinions (Delli Carpini and Keeter 1996). Citizens who are uncertain and have low levels of political interest also tend to avoid mainstream news in favor of more entertaining options such as social media (Prior 2007). Subsequently, social media present a unique environment where people with low levels of political interest and high uncertainty regularly encounter incidental political information. Overall, political interest is used in the operationalization of both relevance and uncertainty and is, therefore, an important psychological factor to account for in the study of agenda-setting effects.

Political interest is also a strong predictor of exposure to political information and a known moderator of media effects resulting from such exposure. Those who have high levels of political interest are more likely to seek out political news and information; however, they are also less susceptible to media influence because their attitudes and opinions are highly crystallized (Delli Carpini and Keeter 1996) and they are more likely to counterargue against inconsistent information (Iyengar, Peters, and Kinder 1982). Conversely, those with low levels of political interest are highly susceptible to media influence, but because they rarely seek out news in favor of more entertaining options, they are unlikely to demonstrate effects from exposure to political information (Converse 1962; Zaller 1992). In forced exposure settings, however, where those with low interest are exposed to political information that they might otherwise avoid, they often demonstrate the largest effects (Arceneaux and Johnson 2013; Iyengar and Kinder 1987).

As a result, incidental political information encountered through social media holds the potential to inform the uninterested and convey the mass media agenda in a novel way that supports the survival of the public agenda despite fragmented audiences. Bulkow, Urban, and Schweiger (2013, 59) write that

... agenda-setting, as it originally was devised, homogenizes the public opinion by bringing in line the judgements of involved and uninvolved persons for issues, which are regarded as important by the media at a certain point in time. Thus, it widens the view of audience members beyond their personal life for problems in the society as a whole.

If social media are able to facilitate an agenda-setting effect, then users who encounter incidental political information will have different perceived issue salience compared with users who do not encounter political information. If social media do not have an agenda-setting effect, then there should be no difference in perceived issue salience between individuals who were exposed to information and those who were not. In addition, the agenda-setting effect should be most prominent among those who report having lower levels of political interest and, therefore, a higher need for orientation. This leads to the following hypotheses:

Hypothesis 1 (H1): Participants exposed to specific political issues through social media are more likely to identify those issues to be important.

Hypothesis 2 (H2): The agenda-setting effect is stronger among politically uninterested participants than among politically interested participants.

Research Design and Method

These hypotheses are tested using an innovative longitudinal experiment that employs the Facebook “Groups” function and a pre-post survey design conducted over seventy-five days in the spring of 2014 ($N = 210$).¹ This design allows one to study the effects of information encountered through Facebook in an unobtrusive way from the user side, leveraging standard features of the platform. Experimental studies of media effects can suffer from concerns about causal claims and external validity because they use unrealistic one-shot exposures, they are administered in unnatural laboratory settings, or they use replicas of news or social media rather than the real thing. The design introduced here is able to overcome some of the limitations inherent to studying media effects through social media.

In this design, the treatment is introduced naturally through the Facebook Groups function, making it so the information appears in the participants’ News Feed

alongside all of the other items shared by their friends and groups. Participants do not have to visit a special website or physical laboratory with this design. In addition, the stories are shared with the participants over a substantial period of time rather than as a one-shot exposure. This is similar to the way in which people normally consume news and information through Facebook and avoids alerting the participants to the content being cued as a variety of stories are shared over the course of the study.

To promote realism in the experiment, participants were recruited under the premise that the purpose of the study was to “explore social networks in undergraduate education.” The Facebook groups that the participants were invited to join were named “UNM [University of New Mexico] Online Community” and “Lobo Online Community” with information shared from the name “UNM News Feeder” using a photo of the university as the profile picture.² The Facebook Groups function allows users to share information with a selected group of people, rather than share it with all of the users they have “friended.” In addition, the Groups function allows researchers to connect with participants without having to “friend” them. This means that researchers never have access to any of the participants’ information that is not already publicly available and helps overcome concerns about privacy. The Groups function also allows administrators to restrict group membership, enabling more control over which participants receive treatment and which do not. Finally, Facebook Groups have a feature detailed below that enables the group administrators to see which group members viewed a post; this allows observation of treatment effects in a more precise way. Overall, there are strong advantages to this design that help to maintain the causal inference that experimental designs offer without sacrificing realism or violating privacy.

Researchers posted stories to the treatment and control group pages on a regular basis over the course of the seventy-five-day experiment. The control group received stories that contained no explicit political valence; for example, “How Are Galaxies Formed?” from BBC³ and “What’s Lost as Handwriting Fades” from the *New York Times*.⁴ The treatment group received hard news stories taken from reputable news sources such as the *New York Times* and the *Washington Post*. To avoid signaling the intent of the study, nine decoy stories (18%) about more general topics such as technology and university-related news were added to the mix of stories shared. In all, fifty-one stories were posted to the treatment group page and fifty-two stories to the control group page. Most stories were shared during the week, with fewer stories shared over the weekend; this equates to sharing about one post per weekday over the course of ten weeks. Only group administrators were allowed to post stories to the group.

This design allows participants to engage with the shared information (or not) at their own discretion. This element is advantageous for promoting realism and improves upon forced exposure designs, however, it also makes it difficult to ensure that the participants are actually exposed to the treatment. For example, participants in the treatment group may not have received treatment because they failed to join the Facebook group, they ignored the posts, or the post was never loaded to the participant's News Feed. It is worth noting that the Facebook Groups feature allows administrators of groups with 250 or fewer members to see which stories shared were "seen by" which members. When a post is shared to the group and loads to a member's Facebook News Feed, it is recorded as "seen by" that member. Although this feature is not a guarantee of treatment, I leveraged the "seen by" feature to develop a robustness test of the findings presented in the paper similar to a test of the treatment on treated. When the analysis is restricted to all participants who saw at least one or more of the posts, and were, therefore, more likely to receive the treatment ("treatment on treated"), all of the reported relationships among those assigned to the treatment group ("intent to treat") hold or become slightly stronger.⁵

Because this project is concerned primarily with whether information on Facebook can serve an agenda-setting function, the hard news stories posted to the treatment group page were limited to four issue areas: immigration, climate change, Ukrainian crisis, and local crime. These issue areas were selected because they represent a mix of local, national, and international issues and because they vary in terms of personal issue relevance and obtrusiveness (Watt, Mazza, and Snyder 1993).

One of the key advantages of a longitudinal design is that it yields panel data that allow for more precise measurement of within-subject treatment effects by controlling for prior levels of the dependent variable (Bartels 2006). To measure agenda-setting effects, a question was included on the pretest and posttest surveys that asked participants to identify three to five issues from a list of nineteen issue areas they felt were most "important in society today" (see Online Appendix A). Ukraine, immigration, climate change, and local crime were included in the list. If Facebook has an agenda-setting function, then participants in the treatment group should be more likely than participants in the control group to name the four cued issues as important.

Participants were recruited from the student population of UNM, a large public university in the Southwest. A randomly selected list of 5,000 registered students was obtained from the University Registrar. Participants were recruited to participate in the study via e-mail sent to their university e-mail addresses and were offered the opportunity to win one of six raffles for \$50 if they completed the

study. Of the 5,000 students recruited into the survey, 781 (16%) responded to the first wave of the survey.⁶ Participants who did not have a Facebook account at the time of the experiment were screened out, resulting in 447 eligible participants who were then randomly assigned to the treatment or control groups.⁷ By the end of the seventy-five-day experiment, 47 percent of the participants were retained in both the treatment and control groups, resulting in a total of 210 participants.

Random assignment of participants to the treatment and control groups should produce balanced populations across all variables in each group. Balance tests indicate that there is balance between the treatment and control groups at the conclusion of the study across those variables that are known to predict study compliance, such as age, sex, race, income, and education (Brehm 1993; see Online Appendix B). However, because the sample is relatively small, a more rigorous equivalence test was also run to identify any significant differences between the groups (see Online Appendix C). The equivalence tests indicate that the treatment and control groups differ significantly only in terms of age; this difference is substantively very small.

The primary dependent variable in this study is a dichotomous variable coded 1 if the participants indicated any of the four cued issues to be important to society in the posttest at time two (T2) and 0 if they did not. To test the hypothesis that news on Facebook has an agenda-setting function (H1), a logistic regression model was used to compare the likelihood that participants assigned to the treatment and control groups indicated the issues cued as important while controlling for whether the participant indicated these issues were important at time one (T1). To test the hypothesis that the agenda-setting effect will be strongest among the least politically interested (H2), the same model was used including an interaction term for political interest at T1 and treatment group assignment. Political interest was measured using a 6-point scale of agreement with the statement, "I am very interested in politics." Political interest was operationalized here as a dichotomous variable, with those who indicated that they "agree" or "strongly agree" with the statement coded as having *high* political interest and all others coded as having *low* political interest. Below, I first present the results of a tabular analysis of the treatment and control groups over time followed by the specific hypothesis tests.

Findings

First, I present a descriptive analysis of change in issue salience over time followed by more rigorous modeling of agenda-setting effects. Figure 1 presents the results of a tabular analysis showing *change* in issue salience

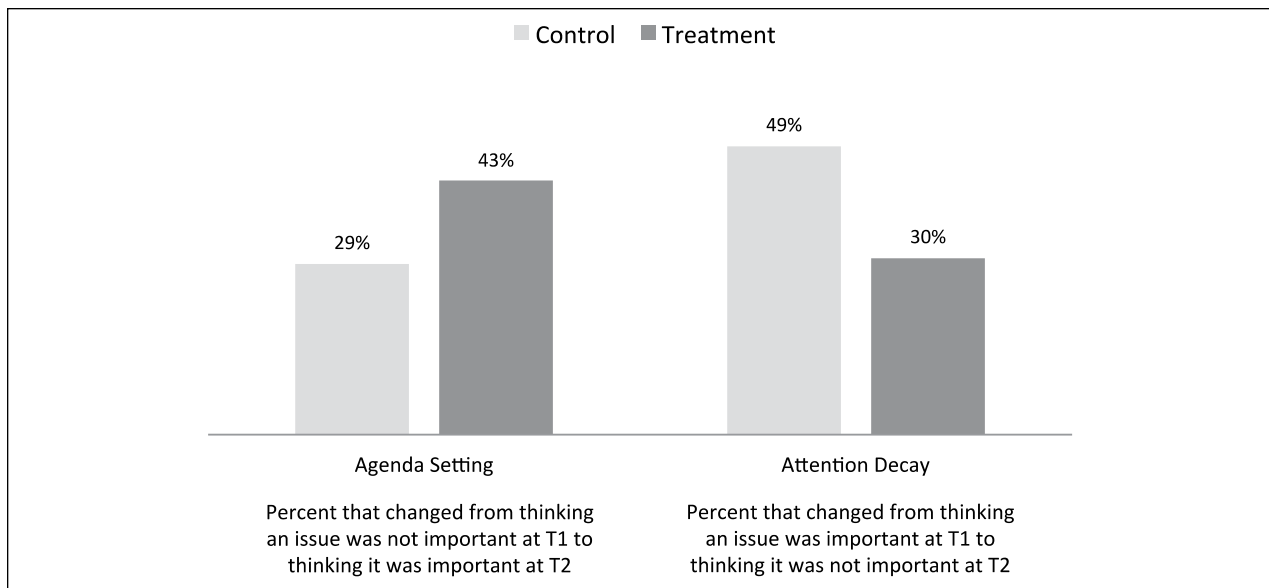


Figure 1. Change in issue salience between T1 and T2.

between T1 and T2 in the treatment and control groups. The figure shows the percentage of those who changed from thinking an issue is not important (“No”) at T1 to thinking an issue is important (“Yes”) at T2 or vice versa (see Online Appendix D for full contingency table). Two interesting trends emerge from this analysis.

First, participants in the treatment group demonstrated overall higher rates of agenda setting (changing from “No” at T1 to “Yes” at T2) compared with the control group. Specifically, 43 percent of the participants in the treatment group and 29 percent of those in the control group changed from thinking none of the issues were important to thinking at least one issue was important over the course of the seventy-five-day experiment. This indicates that sharing political information among the treatment group corresponded with higher rates of agenda-setting effects when compared with the control group.

Second, participants in the treatment group experienced lower levels of attention decay (changing from “Yes” at T1 to “No” at T2) compared with the control group. Over the course of the experiment, 49 percent of those in the control group went from thinking that at least one of the cued issues was important at T1 but not at T2; this was true among only 30 percent of those in the treatment group.⁸ This indicates slowed attention decay in the treatment group as a result of steady exposure to the cued issues.

Attention decay was not specifically anticipated in this study, though some overall effect decay is not necessarily surprising. It may be a result of the relatively unobtrusive nature of the stimulus, the longer than average duration of

the study, or the time lag between the final Facebook post and the posttest.⁹ Compared with other experimental studies of agenda setting, this experiment is substantially longer, and the stimulus is somewhat less overt. For example, Iyengar, Peters, and Kinder (1982) exposed their participants to thirteen to fifteen minutes of TV news footage about an issue over six days. Similarly, Althaus and Tewksbury (2002) exposed their participants to thirty to sixty minutes of online and print newspaper content every day for six days. Perhaps if the present study were shorter, less attention decay would be observed. Alternatively, over the course of the study, participants may have become habituated to the stimulus, where after a period of heightened interest, participants essentially become bored of the issues (Downs 1972; Watt, Mazza, and Snyder 1993). What is important to highlight, however, is that attention decay was lower among the treatment group (30%) than the control group (49%), which suggests that media agenda cueing helped to maintain a basic level of issue salience among the sample.¹⁰

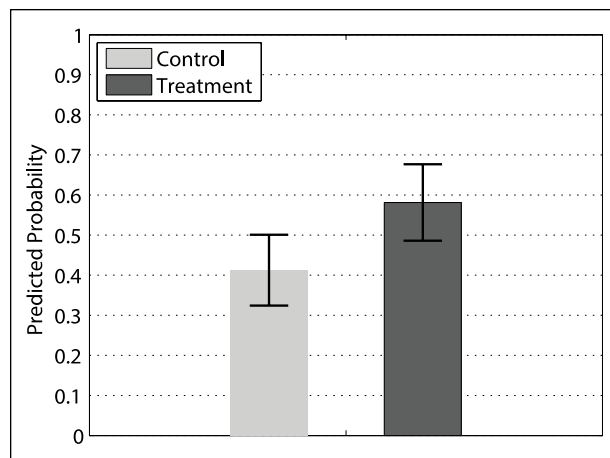
Table 1 presents logistic regression models to more precisely test H1 and H2. Model 1 (M1) in Table 1 shows the results of the logistic regression predicting perceived issue salience at T2 while controlling for issue salience at T1. Corresponding Figure 2 presents these results as the predicted probability of indicating one of the cued issues to be important at T2. Compared with the control group, being in the treatment group increased the probability that participants would perceive the issues cued to be important at T2 by 41 percent (this corresponds to a 17 percentage point difference

Table 1. Logistic Regression of Treatment Effect on Perceived Issue Salience.

	M1	M2
Issue Importance at T1	1.04*** (0.29)	1.01*** (0.32)
Treatment Group	0.73** (0.29)	1.00*** (0.37)
Political Interest		1.14*** (0.44)
Pol. Interest × Treatment		−0.61 (0.69)
Constant	−.96*** (0.26)	−1.32*** (0.34)
N	210	193
χ^2	0.00	0.00
Pseudo- R^2	.07	.09
Log-Likelihood	−135.59	−121.40

Regression coefficient presented with standard errors in parentheses.

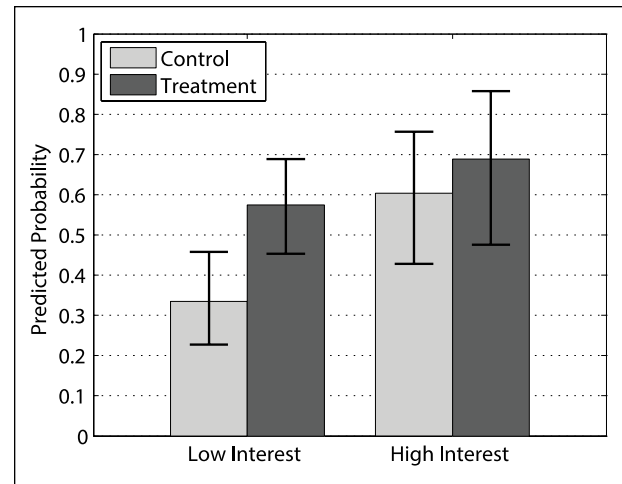
* $p < .10$. ** $p < .05$. *** $p < .01$.

**Figure 2.** Predicted probability of perceiving a cued issue to be important.

Bars show 95% confidence intervals.

between the control and treatment groups). Perceived issue salience at T1 had a strong effect of perceived issue salience at T2; this suggests some stability in issue attitudes, but a stability that can be affected by exposure to information.

The second hypothesis posited that the agenda-setting effect would be stronger for those who are less politically interested. Model 2 (M2) in Table 1 presents the results of the logistic regression predicting issue importance for any of the four issues cued using an interaction term between the experimental condition and level of political interest. While the interaction term was not significant at the $p = .05$ level in the model, postestimation analysis using Clarify showed that the treatment had a significant

**Figure 3.** Predicted probability of perceiving a cued issue to be important by political interest.

Bars show 95% confidence intervals.

effect on perceptions of issue importance for participants with low political interest ($p < .05$) but not a significant effect among participants with high levels of political interest.¹¹ Participants with high political interest in the treatment group were not significantly more likely than participants in the control group to perceive the cued issues as important.

Treatment effects among participants with low political interest were statistically significant and substantively large. Figure 3 shows that the effect of being in the treatment group for those with low political interest increased the predicted probability of selecting one of the cued issues as important by 73 percent (corresponding with a difference of 24 percentage points). For those with high political interest, the predicted probability of thinking one of the cued issues was important was 68 percent in the treatment group and 60 percent in the control group, but this difference was not statistically significant.

The results of M1 in Table 1 support the first hypothesis—that encountering political information through social media can have an agenda-setting effect. The subsequent comparison of high and low interest participants in M2, however, indicates that participants with low levels of political interest were most susceptible to agenda cueing and largely drove the significant findings among the full sample. While this finding does not conform perfectly to the first hypothesis, it does support H2, a hypothesis that was guided by existing literature that suggests media effects are most common among those with lower levels of political interest. Those with lower political interest have a higher need for orientation and are more likely to be affected by incidental exposure to political information via social media.

Table 2. Multivariate Logistic Regression Predicting Individual Issue Salience at T2.

	Immigration		Climate change		Local crime		Ukraine	
Issue Importance at T1	1.93*** (0.40)	1.97*** (0.41)	1.77*** (0.35)	1.70*** (0.36)	1.01* (0.55)	1.09* (0.58)	2.81*** (0.73)	3.05*** (0.82)
Treatment Group	0.51 (0.37)	0.71 (0.48)	0.50 (0.35)	0.59 (0.43)	1.32** (0.55)	2.06** (0.79)	-0.04 (0.73)	1.15 (1.30)
Political Interest		0.83 (0.55)		0.61 (0.51)		0.89 (1.04)		2.66** (1.24)
Interest × Treatment		-0.42 (0.80)		-0.09 (0.74)		-1.60 (1.31)		-2.11 (1.72)
Constant	-2.01*** (0.31)	-2.32*** (0.40)	-1.78*** (0.29)	-1.98*** (0.35)	-3.23*** (0.49)	-3.84*** (0.76)	-3.75*** (0.61)	-5.27*** (1.17)
N	195	193	195	193	195	193	195	193
χ^2	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
Pseudo- R^2	.12	.14	.12	.13	.08	.11	.18	.27
Log-Likelihood	-91.23	-89.54	-101.93	-100.36	-57.47	-53.14	-29.75	-26.53

Regression coefficient presented with standard errors in parentheses.

* $p < .10$. ** $p < .05$. *** $p < .01$.

Building upon the general agenda-setting effect shown in Table 1, Table 2 presents an exploratory multivariate analysis of each cued issue independently. This analysis demonstrates that the treatment effect was not uniform across issue areas. For some issues, one's predispositions are stronger than the treatment effect; for others, the treatment is stronger. For example, concern about the crisis in Ukraine at T1 is a strong predictor of thinking it is important at T2 and the treatment does not have a significant effect on its own. Conversely, concern about local crime at T1 moderately predicts concern about local crime at T2, but the treatment effect in this case is dominant. The differing influence of predispositions at T1 and the treatment effect might be due to varying levels of individual interest in any given issue, or correlated with higher levels of exposure to issue-specific media coverage prior to this study. Future research could leverage a similar research design to this one coupled with national media agenda analysis to further investigate when and under what conditions agenda-setting effects manifest.

The multivariate analysis in Table 2 indicates that two robustness tests are in order. Among the four issues analyzed independently, local crime was the only issue area to register a significant difference between the treatment and control groups ($p < .05$). To be sure that local crime alone is not driving the overall agenda-setting effect, I replicated the analysis removing local crime from the aggregate measure of issue salience (leaving only immigration, climate change, and Ukraine) and found that the results were robust to its omission in both the basic model testing H1 and the interaction model testing H2 (see Online Appendix E).¹² Table 2 also shows that political interest is only statistically significant in the model predicting the issue salience of Ukraine. In response, I

replicated the model omitting Ukraine from the aggregate measure of issue salience at T1 and T2 and found that political interest maintains statistical significance (see Online Appendix F).

Discussion and Conclusion

The postbroadcast media environment presents a world in which we see a shrinking audience for traditional mass media in the face of growing choice in media options and increased selection for the consumer. This environment limits the long-held ability of the mass media to help set the public agenda and promote general agreement about the issues that are important to society. Instead, we see increasing reports of divided and self-contained media ecosystems, some of questionable quality, that limit the ability of the public to accurately assess the world around them and the issues that are important to society as a whole. The findings presented here demonstrate that the information we see in our social media news feed impacts the issues we perceive to be important, and this has known consequences for public opinion and political behavior.

In this study, I tested whether use of social media could convey an agenda-setting effect and provide an alternative pathway for the mass media to reach the public, particularly the inattentive public, using an original experiment run over the course of seventy-five days. At the conclusion, participants in the treatment group exposed to specific political issues were more likely to consider those issues to be important than were participants in the control group. Furthermore, this effect was strongest among those with low levels of political interest who are most likely to avoid political information in

favor of more entertaining options, such as social media. While there is a growing body of excellent research that reexamines traditional effects such as agenda setting and learning as they occur and perhaps differ in the digital media environment (Bode 2016; Conway, Kenski, and Wang 2015; Kobayashi and Ichifuji 2015; Sayre et al. 2010), the findings presented here are among the first to experimentally demonstrate an agenda-setting effect administered through social media.

The longitudinal design of this study enabled examination not only of the psychology of agenda-setting effects by looking at individual differences according to political interest and need for orientation, but also showed how these individual-level differences manifest change in issue salience over time. These findings demonstrate that exposure to political information over the course of the study not only led those in the treatment group to demonstrate agenda-setting effects, but also lessened attention decay over time.

This study further demonstrated that the agenda-setting effect was clearly strongest among those with low political interest who were exposed to incidental political information in the treatment group. Two interesting consequences emerge from this finding. The first is that even though the current media environment allows people with low levels of political interest to easily avoid political news and information, it is likely that social media are able to subvert selective avoidance and convey a modicum of political information to the uninterested. This is generally consistent with prior research by Messing and Westwood (2014) and Anspach (2017), which shows that the social endorsement of news makes people more likely to read news that they might otherwise selectively avoid. Many have raised concern about increased selectivity contributing to an informational divide among citizens, where the information-rich become richer and the information-poor become poorer (Norris 2001). However, I show here that exposure to incidental political information conveyed through social media impacts perceived issue salience, though the degree to which this may help to bridge the divide through deeper learning and understanding remains unclear.

Second, these results suggest that the mass media may not be as marginalized in the postbroadcast environment as previously thought, though their ability to reach the public through social media is heavily mediated by network composition, consumption patterns, and social media algorithms. Bennett and Manheim (2006) argued previously that message targeting and audience selectivity in the current media environment can eliminate the social filtering of political information classically described by Katz and Lazarsfeld (1955) as the “two-step flow of communication.” Media targeting, audience

selectivity, and social isolation, they argue, contribute to a “one-step flow” directly between the recipient and the source. However, in the social media environment, political information is shared socially by members of a network who impart salience and relevance to these stories—similar to how Katz and Lazarsfeld described the role of “opinion leaders” in the traditional broadcast environment. While a one-step flow—from news source directly to recipient—may be more likely today than it was twenty years ago, in a social media setting, the two-step social sharing of information is alive and well. To this point, Turcotte et al. (2015) found that social endorsements on Facebook led people to be more likely to trust shared media content and to seek additional information from the sources of that content especially if it was shared by a perceived opinion leader from their own network. Therefore, as people can easily tailor their individual media diets to suit their wants and needs, traditional agenda-setting effects might be expected to disappear. However, with the increasing prevalence of social media as a source of news, mass media agenda setting may persist to some extent through incidental information shared by sources within our social networks.

There are several limitations to this study that should be noted and hopefully improved upon in future research. One such limitation lies in the use of political interest as the primary contingent condition for agenda-setting effects. Political interest is emphasized here because it is a critical variable to both agenda-setting effects and exposure to incidental information. While political interest is important to consider on its own merits, one’s need for orientation and, therefore, one’s susceptibility to agenda setting can be better measured through more specific techniques (Matthes 2006). Having established here that agenda-setting effects are observable through incidental information exposure on social media, we should move forward seeking to better understand the other conditions under which this effect is more or less likely. Specifically, future studies of this nature should better account for one’s need for orientation by measuring personal and social issue relevance, issue uncertainty, media trust, and source credibility.

Experimental designs are subject to concern about internal and external validity. A series of tests to support the internal validity, including manipulation checks, balance tests, and relying on well-vetted assessments of agenda-setting effects, can lend considerable support to the internal validity of a study such as this one. In terms of external validity, however, the findings may be limited by concern over the generalizability of a student sample (Benz and Meier 2008; Sears 1986). Students tend to be young, and youth may have a higher need for orientation, a characteristic that makes them more susceptible to

agenda-setting effects than members of the general population. Unfortunately, there is little research to date that compares agenda-setting effects among youth versus older generations. One study by Coleman and McCombs (2007) found that people aged eighteen to thirty-four exhibit a high level of correlation between media and personal agendas, though slightly lower than older generations, which the authors attributed to younger populations sourcing their news from the more diverse Internet rather than from traditional media sources. Other research has found that demographics such as age, income, gender, and race are not related to agenda-setting susceptibility; rather, it is psychological factors that matter most (Wanta 1997). It remains unclear whether youth have a higher need for orientation at the individual level compared with older generations as issue relevance and uncertainty vary considerably at all ages. Undoubtedly, future research on the psychological underpinnings of agenda-setting effects should examine youth and their susceptibility to agenda setting in more detail.

In addition, the external validity of the present study may also be threatened by the nature of the research design and, specifically, the source of the news being shared. This experiment demonstrates that an innocuous university-related group can elicit an agenda-setting effect through social media. However, because the source was neither a personal acquaintance nor confirmed opinion leader, concern may be raised that the effect demonstrated is not necessarily a function of social media and the two-step flow of information but is, rather, more similar to that of information shared through a traditional news organization. While this may be true, research shows that stronger social endorsements make people more likely to attend to news (Messing and Westwood 2014). Therefore, news shared by a relatively innocuous source such as the group used here offers a conservative test of agenda setting through social media. One would expect this effect to be stronger if the source were a friend or respected opinion leader who would impart greater personal relevance to a shared story. Future research in this area should take the credibility, opinion leadership, and personal relationship of the source into consideration to better understand the influence of socially shared information.

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Notes

1. This study received approval from the University of New Mexico Institutional Review Board (Study Number 00514).
2. Researchers who use this design should be sure to comply with Facebook user policies.
3. Retrieved from <http://www.bbc.com/news/science-environment-27311552> on November 11, 2014.
4. Retrieved from <http://www.nytimes.com/2014/06/03/science/whats-lost-as-handwriting-fades.html?hp> on November 11, 2014.
5. An equivalence test of the difference between the treatment on treated sample and the control group found that there were no significant differences between these groups and that they were balanced. The mean value for post views among those with high levels of political interest was higher but statistically indistinguishable from those with low political interest in both the control and treatment groups.
6. These individuals are more female, slightly younger, and less white than the general university population. Because I randomly assign participants to treatment and control groups and because I do not expect the effect of agenda setting through social media to be affected by social media users' demographic characteristics, these differences do not present a serious concern (Wanta 1997).
7. This means that about 57 percent of individuals who responded to our survey had a Facebook account. This is lower than Pew's findings that about 70 percent of young people use Facebook (Smith 2014).
8. These results are supported in a logistic regression predicting attention decay between time one (T1) and time two (T2) while controlling for treatment group assignment (Coef. = -1.00, SE = .43, $p = .02$).
9. Participants were allowed up to eleven days to complete the posttest after the final post on Facebook. Additional analysis was conducted controlling for this lag in predicting the agenda-setting effect, but found that the lag had no significant influence on issue salience independently or when interacted with the treatment group, indicating that the findings endure for at least eleven days posttreatment.
10. These results are supported at the 90 percent confidence level in a logistic regression predicting attention decay between T1 and T2 while controlling for political interest (Coef. = -.63, SE = .35, $p = .08$).
11. Michael Tomz, Jason Wittenberg, and Gary King. 2003. "CLARIFY: Software for Interpreting and Presenting

Statistical Results.” *Journal of Statistical Software* 8. Copy at <http://j.mp/k3k0rx>.

12. The fact that local crime has such a strong agenda-setting effect relative to the other issues is not necessarily surprising given the relatively high issue relevance and emerging local scandal with the Albuquerque Police Department at the time of the study, both of which contribute to one’s need for orientation on the issue and, therefore, a stronger effect (McCombs and Reynolds 2009).

Supplemental Materials

Supplemental materials for this article are available with the manuscript on the *Political Research Quarterly (PRQ)* website. Replication data are available at <http://www.jessicafeezell.com/>.

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